The Human Heart and Circulation: Study Notes

1. Why the body needs a circulatory system

Every cell in the body needs a steady supply of oxygen and nutrients, and every cell produces waste such as carbon dioxide that must be carried away. In a tiny organism, these substances can simply diffuse in and out across the surface. In a large animal like a human, most cells are far from the surface, and diffusion alone would be far too slow. The circulatory system solves this problem by pumping blood around the body through a network of tubes, so that every cell is only a short distance from a blood supply.

The human circulatory system has three main parts: the heart, which acts as the pump; the blood vessels, which are the tubes that carry blood; and the blood itself, which is the transport medium. Together they move oxygen, glucose, hormones, antibodies and heat to where they are needed, and carry carbon dioxide, urea and other wastes to the organs that remove them.

2. The double circulatory system

Humans have a double circulatory system. This means that blood passes through the heart twice for every complete circuit of the body. The pulmonary circuit carries blood from the heart to the lungs and back again. In the lungs, blood picks up oxygen and gets rid of carbon dioxide. The systemic circuit carries oxygenated blood from the heart to the rest of the body and brings deoxygenated blood back to the heart.

A double circulation has an important advantage. Blood loses pressure as it passes through the tiny capillaries of the lungs. Because it returns to the heart before going to the body, the heart can pump it again and raise its pressure. This allows blood to reach distant organs quickly and at high pressure, which supports the high metabolic rate of mammals. Fish, by contrast, have a single circulation, where blood passes through the gills and then flows directly on to the body at lower pressure.

3. The structure of the heart

The heart is a muscular organ about the size of a closed fist, found in the chest between the lungs. Its walls are made of cardiac muscle, a special type of muscle that contracts rhythmically without tiring and without needing conscious control. The heart is divided into four chambers. The two upper chambers are the atria, which receive blood returning to the heart. The two lower chambers are the ventricles, which pump blood out of the heart.

The right side of the heart deals with deoxygenated blood and the left side deals with oxygenated blood. A wall of muscle called the septum separates the two sides, so

oxygenated and deoxygenated blood never mix. The wall of the left ventricle is much thicker than the wall of the right ventricle. This is because the left ventricle must pump blood all the way around the body, while the right ventricle only needs to pump blood a short distance to the lungs.

The heart contains valves that keep blood flowing in one direction. The atrioventricular valves sit between each atrium and ventricle. On the right side this is the tricuspid valve and on the left side it is the bicuspid or mitral valve. The semilunar valves sit at the bases of the two main arteries leaving the heart. When a ventricle contracts, the pressure closes the atrioventricular valve, which prevents backflow into the atrium. When the ventricle relaxes, the semilunar valve closes, which prevents blood flowing back from the artery. The closing of the valves causes the familiar lub-dub sound of a heartbeat.

The heart muscle itself needs a supply of oxygen. It is supplied by the coronary arteries, which branch off the aorta and spread across the surface of the heart. If a coronary artery becomes blocked, for example by a build-up of fatty material, part of the heart muscle is starved of oxygen. This can cause a heart attack.

4. The route of blood through the heart

Deoxygenated blood from the body returns to the heart through two large veins called the venae cavae. It enters the right atrium. The right atrium contracts and pushes blood through the tricuspid valve into the right ventricle. The right ventricle then contracts and pumps the blood through a semilunar valve into the pulmonary artery, which carries it to the lungs.

In the lungs, the blood is oxygenated and returns to the heart through the pulmonary veins. It enters the left atrium, which contracts and pushes the blood through the bicuspid valve into the left ventricle. The thick muscular left ventricle contracts strongly and pumps the blood through a semilunar valve into the aorta, the largest artery in the body. From the aorta, blood is carried to every organ except the lungs.

A useful way to remember the order is: body, vena cava, right atrium, right ventricle, pulmonary artery, lungs, pulmonary vein, left atrium, left ventricle, aorta, body.
Notice that the pulmonary artery is the only artery that carries deoxygenated blood, and the pulmonary vein is the only vein that carries oxygenated blood.

5. The cardiac cycle and heart rate

The sequence of events in one heartbeat is called the cardiac cycle. It has three main stages. In atrial systole, the atria contract and push blood into the ventricles. In ventricular systole, the ventricles contract, the atrioventricular valves close, and blood is forced into the arteries. In diastole, the whole heart relaxes and the chambers

fill with blood again from the veins.

The heartbeat is started by a small group of cells in the wall of the right atrium called the sinoatrial node, often called the pacemaker. It sends out regular electrical impulses that spread across the atria and make them contract. The impulse then reaches the atrioventricular node, which delays it slightly so that the atria finish emptying before the ventricles contract. The impulse then travels down special fibres in the septum and spreads up through the ventricle walls from the bottom, squeezing blood upwards into the arteries.

A resting adult heart beats about 60 to 80 times per minute. During exercise, the heart rate increases so that more oxygen and glucose reach the working muscles and more carbon dioxide is removed. Cardiac output is the volume of blood pumped by one ventricle per minute. It is calculated by multiplying the heart rate by the stroke volume, which is the volume of blood pumped in one beat. Regular exercise strengthens the heart muscle, increasing stroke volume, which is why trained athletes often have a low resting heart rate.

6. Blood vessels

There are three main types of blood vessel. Arteries carry blood away from the heart. Because the blood in them is at high pressure, arteries have thick walls containing muscle and elastic fibres. The elastic fibres stretch as each surge of blood passes and then recoil, which smooths out the flow. This surge can be felt as a pulse where an artery passes close to the skin, such as at the wrist or neck.

Veins carry blood back towards the heart. The blood in veins is at low pressure, so veins have thinner walls and a wider lumen, which is the space inside the vessel. Veins contain valves that stop blood flowing backwards. Many veins run between skeletal muscles, and when these muscles contract they squeeze the veins and help push blood back towards the heart.

Capillaries are the smallest blood vessels. They connect arteries to veins and form dense networks called capillary beds in every tissue. The wall of a capillary is only one cell thick, which makes the diffusion distance very short. Oxygen and glucose diffuse out of the blood into the cells, and carbon dioxide and other wastes diffuse from the cells into the blood. Because there are so many capillaries, they provide a huge surface area for exchange.

7. Blood

Blood is made of a pale yellow liquid called plasma, in which several types of cell are suspended. Plasma is mostly water and carries dissolved substances such as glucose, amino acids, carbon dioxide, urea, hormones and proteins. It also carries heat around

the body.

Red blood cells carry oxygen. They contain the red protein haemoglobin, which combines with oxygen in the lungs to form oxyhaemoglobin and releases it in the tissues. Red blood cells are adapted for their job: they have no nucleus, which leaves more room for haemoglobin, and they have a biconcave disc shape, which gives them a large surface area for oxygen to diffuse in and out.

White blood cells are part of the immune system. Some, called phagocytes, engulf and digest pathogens. Others, called lymphocytes, produce antibodies that attach to specific pathogens and help destroy them. Platelets are small fragments of cells that help the blood to clot at the site of a wound, which stops bleeding and prevents pathogens from entering the body.

8. Keeping the circulatory system healthy

Coronary heart disease happens when fatty deposits build up in the walls of the coronary arteries, narrowing them and reducing the flow of blood to the heart muscle. Risk factors include smoking, a diet high in saturated fat, high blood pressure, lack of exercise, and a family history of heart disease. Some of these factors, such as diet, smoking and exercise, can be changed by the choices a person makes.

Treatments for heart disease include medicines called statins, which lower cholesterol in the blood, and stents, which are small mesh tubes inserted into a narrowed artery to hold it open. In more serious cases a surgeon may perform a bypass operation, using a section of blood vessel from elsewhere in the body to carry blood around the blocked part of the artery. Faulty heart valves can be replaced with mechanical valves or with valves taken from animals.

Summary: the heart is a double pump with four chambers and one-way valves. It drives blood through arteries, capillaries and veins in two circuits, one to the lungs and one to the body. Blood carries oxygen in red blood cells, fights infection with white blood cells, clots with platelets, and transports dissolved substances in plasma.